How We Calibrated WOFOST to Texcoco Farm Yields

What was done, why, and how to read the results

Gradient Descent Farmers — companion to the technical report

October 19, 2026

1. In one paragraph

Before calibration, the crop simulator predicted yields 2.5 to 5 times higher than what Texcoco farmers report to the Mexican government (6.5 vs 2.6 t/ha for maize, 6.1 vs 1.8 for barley, 9.8 vs 1.9 for faba bean). We picked a few model settings ("knobs") with a clear physical meaning and tried many combinations. For each crop we kept the combination whose simulated yields came closest to the observed yields of 2006–2021, and we checked it on 2022–2025. After calibration the simulated averages are within 0.06–0.25 t/ha of the observed ones in the fit years. But the calibrated model does **not** predict individual years better than simply using the observed 2006–2021 average, and for maize it does worse. Calibrated yields are realistic *levels* for scenarios; they are not forecasts. A second finding matters for the rotation work: once calibrated, maize no longer yields more after barley or faba bean than after maize.

This is the second version of the calibration. An independent review of the first version found that it compared maize with the wrong mix of rotations and contradicted a local fertilizer trial; Section 7.2 explains what changed.

2. What "calibration" means here

WOFOST, the crop model we use, does not learn from yields. It computes a yield day by day from weather, soil and about a hundred crop parameters (see the companion *What Kind of Model Is WOFOST?*). Those parameters come from field experiments in Europe: they describe modern varieties grown with good weed, pest and fertilizer management. Texcoco's rainfed farms are different, so the model's yields are attainable yields, not farm yields.

Calibration means adjusting a small number of those parameters so that the model's output matches observed data for the place we care about. It is a curve fit, but of a few physically meaningful numbers inside the model, not of the yield itself.

Why not just multiply the yields by a correction factor? The farmer notebook did exactly that: it multiplied every maize yield by about 0.4, barley by 0.28 and faba bean by 0.18. That fixes the average but nothing else:

- the model still believes fertilizer, frost and drought act on a 6–10 t/ha crop, so the size of every scenario effect ("20% less rain", "no fertilizer") is wrong;
- the same factor is applied in every year, whatever happened in that year.

Changing parameters *inside* the model lets the lower yield interact with water, nitrogen and frost the way a real low-input crop would.

3. The ingredients

3.1 Observed yields: SIAP

SIAP (Servicio de Información Agroalimentaria y Pesquera) publishes, every year and for every municipality, the area sown, the area harvested, the area lost and the production of each crop. We keep the rows for Texcoco (state 15, municipality 99), the spring–summer cycle and rainfed land (`farm_tool.siap_texcoco`). From them we use the **yield per sown hectare**:

$$y^{\text{obs}} = \frac{\text{production (t)}}{\text{sown area (ha)}}. \tag{1}$$

Dividing by the *sown* area (not the harvested area) counts fields that were lost to frost or drought as zero yield. That is what a failed season looks like in the model, so the two are comparable. In 2011 and 2019, for example, 51% and 64% of the maize area was lost, and the yield per sown hectare falls to 1.0 and 0.8 t/ha.

Crop	Years with data	Mean 2006–2021	Mean 2022–2025	Sown area, 2022–2025
Maize	2005–2025	2.56 (16 years)	2.98	900–1,400 ha
Barley	2008–2025	1.78 (14 years)	1.55	4–9 ha
Faba bean	2005–2019, 2022–2025	1.94 (14 years)	1.17	10–12 ha

Table 1: The observations used (yields in t/ha per sown hectare). Barley has no data before 2008 and faba bean none for 2020–2021.

3.2 Simulated yields, combined as SIAP would see them

The simulator (`wofost_rotations.py`) runs four rotations (maize every year; maize–barley; maize–faba bean; maize–barley–faba bean) from 2005 to 2025. Every *phase* of a rotation is simulated: maize–barley is run once starting with maize and once starting with barley, so every crop appears every year. Seasons are chained, so the water and nitrogen left by one crop start the next one. The yield we compare is the **water- and nitrogen-limited yield** (rainfed, with fertilizer, with the grain-N constraint), the most realistic of the three production levels the simulator computes. The first year, 2005, is a "spin-up" that starts from assumed soil water and nitrogen, so it is left out.

SIAP reports one number per crop and year for the whole municipality, so the simulated seasons must be combined into one number in the same way. For maize, the share of fields following each crop matters a lot, and SIAP tells us roughly what it is. Maize after barley can cover at most the barley area of the year before, and the same holds for faba bean:

$$w_t^{\text{barley}} = \frac{\text{barley area}_{t-1}}{\text{maize area}_t}, \qquad w_t^{\text{faba}} = \frac{\text{faba bean area}_{t-1}}{\text{maize area}_t}, \qquad w_t^{\text{maize}} = 1 - w_t^{\text{barley}} - w_t^{\text{faba}}, \tag{2}$$

and the simulated maize yield of year t is the weighted average of maize after each crop:

$$y^{\text{sim}}_{\text{maize},t} = \sum_{p \,\in\, \{\text{maize, barley, faba}\}} w_t^{p}\ \bar{Y}_{\text{maize after } p,\ t}. \tag{3}$$

Over 2006–2021 barley and faba bean areas are only 7.4% of the next year's maize area on average (at most 24.8% in 2009, and under 3% since 2019; in 2023, 16 ha against 1,390 ha). So almost all Texcoco maize follows maize. The rest of the area is assumed to follow maize too: SIAP has no field histories, and other preceding crops (oats, for example) are not simulated. Barley and faba bean are averaged over all rotations and phases, since SIAP cannot tell what they follow. This is `farm_tool.simulated_like_siap`, used by the calibration and by `farm_gap` in the notebook.

Why this matters. The first version of the calibration averaged maize over the four rotations equally, as if three quarters of the fields followed barley or faba bean. Re-weighted by the real areas, its calibrated maize fell from 2.35 to 1.97 t/ha in the fit years and to 1.90 in 2022–2025 (holdout RMSE 2.27 instead of 0.89). Its good holdout score depended on the wrong weighting.

3.3 Fit years and holdout years

A model with enough knobs can match any short series, including its noise, and then fail on new years. This is called *overfitting*. To check for it, the years are split:

- **Calibration (fit) years, 2006–2021**: the only years the search looks at when choosing knob values.
- **Holdout years, 2022–2025**: never used to choose knob values. Four years is little, so the holdout tells us about the level more than about year-to-year agreement.

The holdout is not pristine, though: while designing the first version we looked closely at 2023 (Section 5) and at diagnostics that printed holdout scores. The diagnostics now score fit years only.

4. How a fit is scored, and what it is compared with

For one crop and the n years with an observation, let $e_t = y_t^{\text{sim}} - y_t^{\text{obs}}$ be the error in year t. We report:

$$\text{Level:} \quad \bar{y}^{\text{sim}} \text{ vs } \bar{y}^{\text{obs}} \quad \text{(the two means; their ratio is the old correction factor)} \tag{4}$$

$$\text{Bias:} \quad \bar{e} = \frac{1}{n}\sum_t e_t \quad \text{(positive = model too high)} \tag{5}$$

$$\text{RMSE:} \quad \text{RMSE} = \sqrt{\frac{1}{n}\sum_t e_t^2} \quad \text{(typical size of the error, in t/ha)} \tag{6}$$

$$\text{Correlation:} \quad r = \frac{\sum_t (y_t^{\text{sim}} - \bar{y}^{\text{sim}})(y_t^{\text{obs}} - \bar{y}^{\text{obs}})}{\sqrt{\sum_t (y_t^{\text{sim}} - \bar{y}^{\text{sim}})^2}\sqrt{\sum_t (y_t^{\text{obs}} - \bar{y}^{\text{obs}})^2}} \tag{7}$$

r runs from -1 to 1. It measures whether good and bad years line up (1 = perfectly, 0 = no relation), whatever the level. With 16 years, r must be above about 0.5 to be distinguishable from zero.

The search minimizes RMSE. It is useful to know that

$$\text{RMSE}^2 = \bar{e}^{\,2} + \text{SD}(e)^2, \tag{8}$$

so RMSE punishes both a wrong level ($\bar{e}$) and wrong year-to-year swings (SD(e)). When the model's swings do not line up with the observed ones, the cheapest way to lower RMSE is to make the model swing *less*, or to settle slightly below the observed level if higher yields come with bigger swings.

Naive baselines. A model is only useful if it beats a trivial prediction. We compare every result with two of them, both computed from SIAP alone:

- **Climatology**: every year gets the 2006–2021 SIAP average of that crop.
- **Persistence**: every year gets last year's SIAP yield (years whose previous year has no observation are left out).

The **skill score** $1 - \text{RMSE} / \text{RMSE}_{\text{climatology}}$ is positive when the model beats climatology, zero when it ties and negative when it does worse.

5. Diagnosis: why is the default model so far off?

Before fitting anything, we changed one assumption at a time and reran all rotations (`python -m src.crop_models.calibration --sensitivity`). Table 2 summarizes the results.

Change from the default	Maize	Barley	Faba bean	Maize r	Maize ended by frost
None (default)	6.47	6.15	9.79	0.35	69%
Phenology ×0.8 (shorter cycle)	9.41	5.56	9.80	0.39	25%
Phenology ×0.65	10.64	4.24	8.22	0.40	6%
Photosynthesis ×0.5	3.41	2.77	2.53	−0.02	69%
Fertilizer N recovery 0.25 (not 0.5)	4.16	4.97	9.77	0.39	69%
Soil N mineralization halved	4.60	3.83	9.74	0.38	69%
Frost kills at −2 °C (not 0 °C)	6.79	6.15	9.89	0.35	44%
Soil 50 cm deep (not 100 cm)	6.06	6.16	9.79	0.35	69%
SIAP (observed)	2.56	1.78	1.94		

Table 2: One-at-a-time sensitivity: mean simulated yield 2006–2021 (t/ha) over the years with SIAP data, maize weighted by SIAP areas.

What the table and the model's internal variables tell us:

1. **The crop parameters describe high-yield European crops.** Without water or nitrogen limits ("potential" yield) the default simulates 11.2 t/ha of maize, 12.0 of barley and 10.0 of faba bean. Rain barely matters on average: the 100 cm soil stores 236 mm of plant-available water and May–October brings about 770 mm on average, so the rainfed yields are the same as the potential ones.

2. **Nitrogen is the only brake in the default model, and only on maize and barley.** Default maize takes up about 150 kg of N/ha: 80 kg from fertilizer (160 kg/ha applied, of which the model lets the crop recover half) and about 64 kg per season from the soil's own organic matter (mineralization). Halving either source lowers maize by about 2 t/ha. Faba bean is a legume that takes 80% of its nitrogen from the air, so nothing reduces it.

3. **Phenology and frost.** NASA POWER weather comes from a grid cell whose average elevation is 2,633 m, about 380 m above the fields at El Batán. Air cools about 6.5 °C per km of altitude, so the weather the model sees is roughly 2.5 °C colder than the fields. Maize development is driven by accumulated temperature, so it is slow. The default maize flowers about 87 days after sowing (plausible for Valles Altos hybrids) but then needs about 100 more days to fill its grain. In 69% of seasons the first frost (a night at or below 0 °C) ends the crop before it matures; when it matures, the median date is December 5.

4. **Frost threshold and soil depth matter little.** Neither changes the mean by more than 0.4 t/ha. (The "frost kills at −2 °C" row means *fewer* frosts; the `frost_minus2` scenario of the technical report means colder nights, i.e. *more* frosts.)

5. **Photosynthesis is a strong level lever but erases the year pattern.** Halving it brings all three crops much closer to SIAP, but maize's correlation drops from 0.35 to about 0.

Why does continuous maize almost fail in 2023 in the default model? SIAP reports 2023 as Texcoco's best maize year (4.0 t/ha). In the default model, maize after barley or faba bean does well in 2023 (11–13 t/ha), but maize after maize nearly fails (0.01 t/ha):

1. the 2022 maize was sown late (June 15) and kept transpiring until January 5, 2023;
2. on March 1, 2023 the soil therefore held under 10 mm of water the plant could use, against 130–200 mm when barley or faba bean had been grown in 2022 (no rain falls in the dry season to refill it);

3. the 2023 crop was sown on May 15, after about 24 mm of early-May rain, and then June was almost rainless (a 28-day dry spell; 2023 had the driest May–October of 2005–2025 in the POWER data, 478 mm);
4. with no water to take up, the seedlings could not grow; their leaf area shrank from 0.05 to 0.003 and never recovered. WOFOST has no "re-sowing", which a farmer would do.

Because almost all Texcoco maize follows maize (Section 3.2), this failure dominated the default model's 2023. The calibrated model's shorter maize cycle (chosen on 2006–2021 only) avoids it: calibrated continuous maize gives 2.0 t/ha in 2023.

6. The knobs, and the fertilizer-trial constraint

We use four knobs, each tied to a mechanism found in Section 5. The defaults reproduce the uncalibrated model exactly.

Knob	WOFOST parameters changed	Default	What it represents
Phenology scale s	TSUM1, TSUM2 $\times s$	1	A shorter or longer-cycle variety
Photosynthesis scale p	AMAX_REF, AMAX_SLP, EFFTB $\times p$	1	Losses WOFOST does not simulate: weeds, pests, diseases, uneven plant stands, landrace genetics
Fertilizer N recovery f	share of each N dose the crop can access	0.5	Lower doses than the 160 kg/ha plan, losses to leaching and volatilization
Soil mineralization m	N released per season $= m\times$ soil total N	0.5%	Fertility of the soil organic matter (one value for the site)

Table 3: The calibration knobs. The first three are set per crop; mineralization is shared by all crops.

Phenology (TSUM). WOFOST tracks a development stage D that goes from 0 at emergence to 1 at flowering and 2 at maturity. Each day it grows by the day's effective temperature divided by a thermal-time requirement:

$$D_{t+1} = D_t + \frac{T_{\text{eff}}(t)}{\text{TSUM1}} \ (D < 1), \qquad D_{t+1} = D_t + \frac{T_{\text{eff}}(t)}{\text{TSUM2}} \ (D \geq 1), \tag{9}$$

where, for maize, $T_{\text{eff}} = \min(\max(T - 6, 0), 24)$ in °C day. Multiplying TSUM1 and TSUM2 by $s < 1$ makes every stage arrive sooner. Table 4 shows what this means in days after sowing at Texcoco. Phenology depends only on the sowing date and temperature, so it is the same in every rotation.

Crop	Phenology scale	Days to flowering	Days to end of season	Seasons reaching maturity
Maize	1.0 (default)	87	191	31%
	0.8 (calibrated)	72	155	75%
Barley	1.0 (default)	63	117	88%
	0.9 (calibrated)	57	107	100%
Faba bean	1.0 (default)	57	145	75%
	0.9 (calibrated)	52	130	88%

Table 4: Simulated phenology, medians over the 16 seasons of 2006–2021 (median sowing: maize May 27, barley July 1, faba bean June 16). A season ends at maturity, at the first frost, or on February 28.

Photosynthesis. In WOFOST 8.1 the maximum CO_2 uptake of a leaf depends on its nitrogen content: $A_{\max} = \text{CO2 factor} \times \text{temperature factor} \times \min(\text{AMAX_REF},\ \text{AMAX_SLP} \cdot \text{SLN})$, and EFF is the light-use efficiency of a leaf in dim light. Scaling all three by p scales each leaf's CO_2 uptake, and so the crop's daily sugar production, by p (except where $A_{\max}$ falls below 2 kg/ha/h, where the model holds it at 2). Everything downstream (leaf growth, roots, grain, N demand) responds to it, which is why this works better than multiplying the final yield.

Nitrogen. Each fertilizer dose of F kg N/ha makes $f \cdot F$ available to the crop; the rest is lost. The soil also releases $m \times 12{,}709$ kg N/ha (IGBP total N in 0–100 cm) per season, spread evenly over the first 150 days from March 1. With the defaults that is $0.005 \times 12{,}709 \approx 64$ kg.

The fertilizer-trial constraint. SIAP tells us how much a field yields, not where its nitrogen came from. The first version of the calibration lowered maize yields by cutting nitrogen (fertilizer recovery 0.2 and half the mineralization). Its unfertilized maize then fell to 0.29 t/ha against 2.35 fertilized, a ratio of 0.12. The only local nitrogen data in the repository say the opposite: in a rainfed hybrid trial at INIFAP's Texcoco station, maize gave 5.2 t/ha without fertilizer and 5.6 t/ha with 160 kg N/ha, a ratio of 0.93 (Alonso-Sánchez et al. 2023, doi:10.28940/terra.v41i0.1577). So maize candidates are now also run without fertilizer, and only those whose continuous-maize ratio

$$\rho = \frac{\text{mean yield without fertilizer, 2006–2021}}{\text{mean yield with fertilizer, 2006–2021}} \tag{10}$$

is within 0.1 of 0.93 are eligible (`within_trial`). The trial is one research-station experiment, so applying it to the whole municipality is an assumption; the tolerance of 0.1 reflects that.

How the knobs reach the model. The calibration is passed to `simulate_rotations(..., calibration=...)` as a dictionary. Crop parameters are applied as PCSE *overrides* on the parameter provider of each season. (Editing the crop parameter dictionary directly does not work: PCSE reloads the variety's parameters when the crop is sown.) When no calibration is passed, the code takes exactly the old path: a rerun of the default baseline, `no_fertilizer` and `canicula50` reproduces the saved CSVs exactly.

7. The search

7.1 Grid search, one crop at a time

A **grid search** lists a few values for each knob and simulates every combination (a "candidate"). For each candidate we simulate the rotations that contain the crop, compute the crop's RMSE over 2006–2021, and keep the eligible candidate with the lowest RMSE. Figure 1 shows the loop.

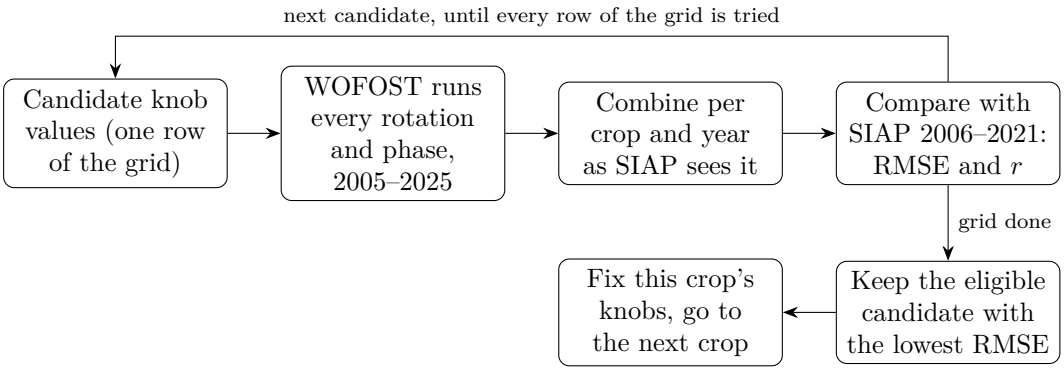


Figure 1: The calibration loop. Maize candidates are also run without fertilizer to check the trial constraint. The holdout years 2022–2025 never enter it.

Trying every combination of all knobs for all three crops at once would take days, so the crops are fitted one at a time ("coordinate search"):

1. **Barley** (15 candidates): phenology $\{0.8, 0.9, 1\}$, photosynthesis $\{0.4, 0.425, 0.45, 0.475, 0.5\}$, N recovery 0.5. The fine photosynthesis steps matter: barley's level climbs about 0.3 t/ha per step.
2. **Faba bean** (15): the same grid. Its N recovery barely matters because it fixes its own N.
3. **Maize** (88), with barley and faba bean at their fitted values: phenology $\{0.8, 1\}$, photosynthesis $\{0.35, 0.4, 0.45, 0.46, 0.47, 0.48, 0.49, 0.5, 0.6, 0.75, 1\}$, and four nitrogen supplies (fertilizer recovery, mineralization): (0.2, 0.25%), the first version's choice; (0.5, 0.5%), the defaults; (0.5, 0.75%); and (0.5, 1%). Each maize candidate is also run without fertilizer.

Maize goes last because it depends on its partners (it inherits the water and nitrogen they leave) more than they depend on it. To save time, the search only simulates the water- and N-limited level. The search takes about 13 minutes on 8 processor cores; the maize photosynthesis values 0.46–0.49 were added in a second pass after the first pass showed the level jumping from 2.0 to 2.8 t/ha between 0.45 and 0.5 (Figure 4). `fit_history.csv` holds every candidate of both passes.

Ties and grid edges. RMSE and r are compared to three decimals; equal RMSE goes to the higher correlation, and then to the candidate closest to the defaults, measured as $\sum_k |v_k/\text{default}_k - 1|$. The search also reports choices on the edge of the grid, since the best value may lie outside it.

7.2 How the calibration got here

Version	What the search chose	What we learned and changed
1a	Maize phenology ×0.6, photosynthesis ×0.6, N recovery 0.25: all on the edge of the grid	×0.6 means flowering 56 days after sowing, a variety not grown in Texcoco. Phenology was limited to ×0.8–1.
1b–1d	Maize at default phenology and photosynthesis, with less and less N: recovery 0.15, then 0.05, then 0.2 with half the mineralization (mean 2.35, RMSE 0.71, r 0.39)	Published as the first calibration.
Review	An independent review found that (i) maize was averaged over rotations equally, though almost all Texcoco maize follows maize; (ii) the model did not beat the SIAP average; (iii) the N choice sat on the edge of the grid and contradicted the Texcoco fertilizer trial; (iv) barley's ×0.8 phenology was an artifact of a coarse grid	Maize is now weighted by SIAP areas; baselines are reported; maize candidates must keep the trial's fertilizer response, may lower photosynthesis, and can have more soil N; barley and faba bean get finer photosynthesis steps; diagnostics score fit years only.
2 (current)	Maize phenology ×0.8, photosynthesis ×0.47, default N supply; barley and faba bean phenology ×0.9, photosynthesis ×0.475	See Section 8.

Table 5: The versions of the calibration. A choice on the edge of the grid means the best value may lie outside it.

Two limits are deliberate and come from agronomy, not from the fit:

- **Phenology at least** ×0.8, to keep crop calendars realistic. Maize sits on this bound.
- **Photosynthesis at least about** ×0.35. Below that the crops barely establish.

Maize's N supply is also reported as "on the edge", but that is a tie: once photosynthesis is low enough to match the farm level, nitrogen no longer limits maize, so the default and higher N supplies give the same yields and the tie goes to the defaults.

8. Results

Crop	Phenology	Photosynthesis	Fertilizer N recovery
Maize	×0.8	×0.47	0.5 (default)
Barley	×0.9	×0.475	0.5 (default)
Faba bean	×0.9	×0.475	0.5 (default)
Site	Soil mineralization 0.5% of total N per season (default, about 64 kg N/ha)		

Table 6: The fitted knobs, stored as `CALIBRATED` in `src/crop_models/calibration.py`. 56 of the 88 maize candidates passed the trial constraint; the chosen one has a fertilizer ratio of 0.995.

		Fit years 2006–2021				Holdout 2022–2025		
Crop	Prediction	sim / obs (t/ha)	RMSE	r	skill	sim / obs (t/ha)	RMSE	skill
Maize	default	6.47 / 2.56	4.49	0.35	−5.4	5.50 / 2.98	4.49	−4.6
	calibrated	2.31 / 2.56	1.07	0.11	−0.53	2.35 / 2.98	1.03	−0.29
	climatology	2.56 / 2.56	0.70		0	2.56 / 2.98	0.80	0
	persistence	2.55 / 2.56	0.97	0.04	−0.39	2.93 / 2.98	1.25	−0.57
Barley	default	6.15 / 1.78	4.37	−0.08	−22.8	5.81 / 1.55	4.28	−17.4
	calibrated	1.84 / 1.78	0.25	−0.03	−0.37	1.55 / 1.55	0.21	0.08
	climatology	1.78 / 1.78	0.18		0	1.78 / 1.55	0.23	0
	persistence	1.80 / 1.76	0.17	0.58	0.10	1.53 / 1.55	0.05	0.80
Faba bean	default	9.79 / 1.94	7.87	−0.11	−15.0	9.59 / 1.17	8.45	−9.9
	calibrated	1.75 / 1.94	0.61	−0.17	−0.23	1.78 / 1.17	0.62	0.20
	climatology	1.94 / 1.94	0.49		0	1.94 / 1.17	0.78	0
	persistence	2.00 / 1.94	0.38	0.73	0.23	1.16 / 1.19	0.08	0.90

Table 7: Calibrated WOFOST against the default model and the two naive baselines. Skill = $1 - \mathrm{RMSE} / \mathrm{RMSE}_{\mathrm{climatology}}$; positive beats the SIAP average. Persistence uses one year fewer for barley (fit) and faba bean (holdout). Holdout correlations rest on four years and are not shown.

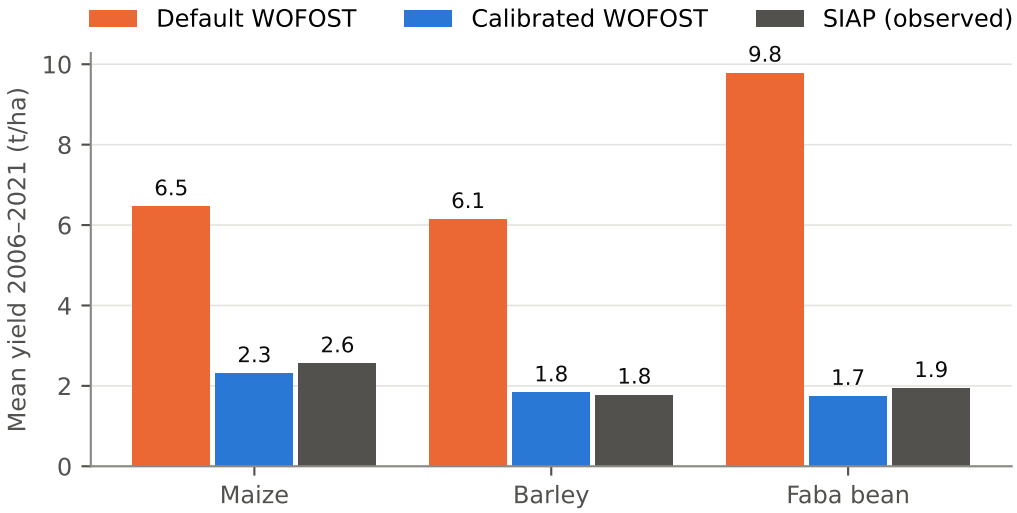


Figure 2: Mean yield over the fit years: the calibrated model (blue) now sits next to SIAP (gray).

How to read the results.

- **The level is fixed.** In the fit years the calibrated means are within 0.06 (barley), 0.20 (faba bean) and 0.25 (maize) t/ha of SIAP. The `farm_gap` correction factors over 2006–2025 go from 0.42, 0.28 and 0.18 to 1.14, 0.98 and 1.01.

- **The calibrated model does not beat the SIAP average.** In the fit years, climatology has a lower RMSE than the calibrated model for every crop. In the holdout, the calibrated model only edges out climatology for barley and faba bean, and persistence beats both by far: those SIAP series barely change from one year to the next. For maize, the calibrated model is worse than climatology in both periods (skill −0.53 and −0.29).

- **Year-to-year agreement is poor.** Calibrated maize has $r = 0.11$ in the fit years. Its swings are about as large as the observed ones (standard deviation 0.88 vs 0.72 t/ha) but fall in the wrong years (Figure 3): it reproduces the 2011 loss but also invents losses in 2010, 2012 and 2020 and misses the 2019 loss. Faba bean is too steady (0.23 vs 0.51 t/ha) and does not follow

SIAP's decline from 2.6 t/ha in 2005–2009 to about 1.2 in recent years, hence its holdout bias of +0.6 t/ha.

- **Maize sits on a steep slope.** Calibrated maize yields 2.0 t/ha at photosynthesis ×0.45 and 2.8 at ×0.5 (Figure 4). Small changes in the year's conditions push it across that slope, which is part of why its year pattern is erratic.
- **2023.** Calibrated maize gives 2.0 t/ha in 2023 in every rotation, against SIAP's 4.0.

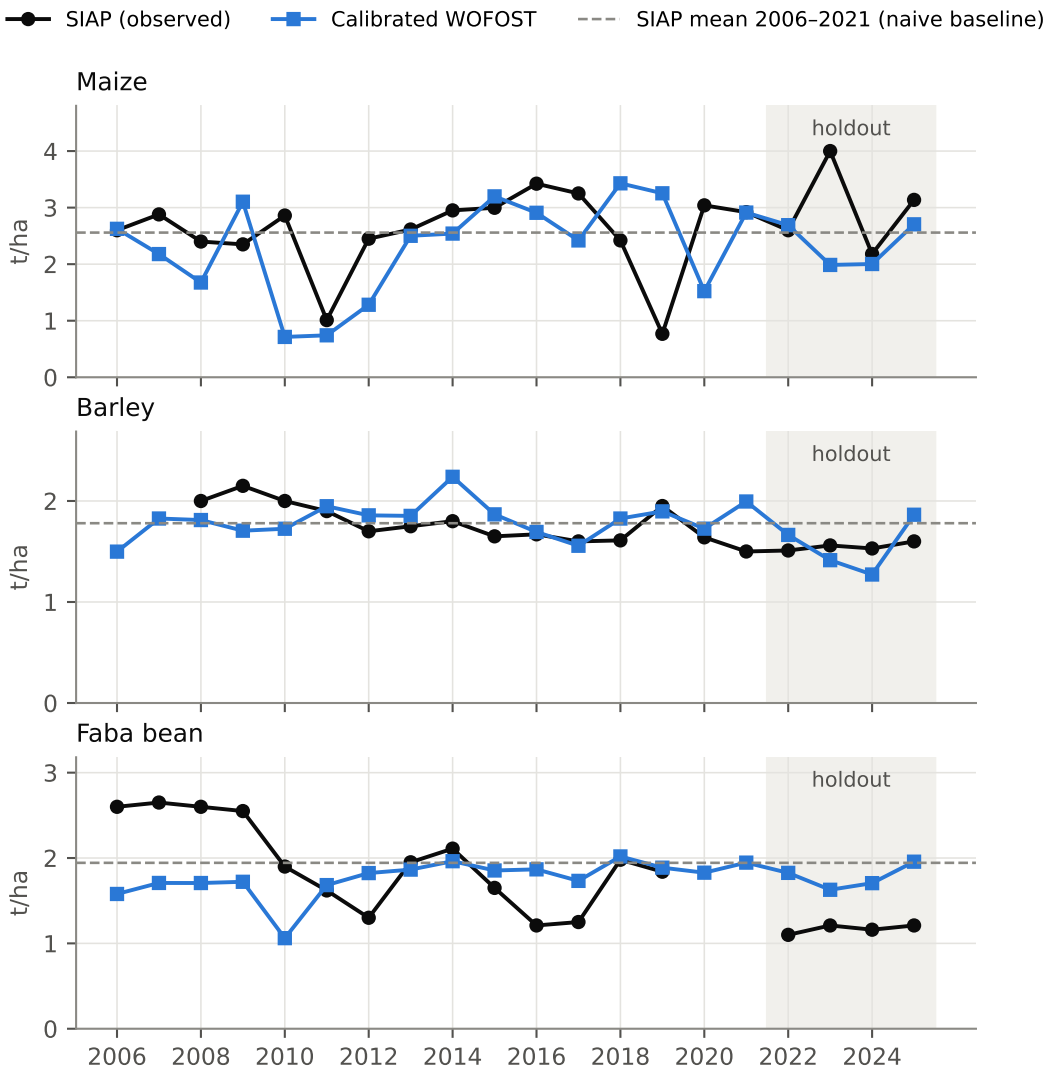


Figure 3: Year by year: calibrated WOFOST (blue squares) against SIAP (black dots) and the climatology baseline (dashed). The shaded years are the holdout. Missing black dots are years without SIAP data.

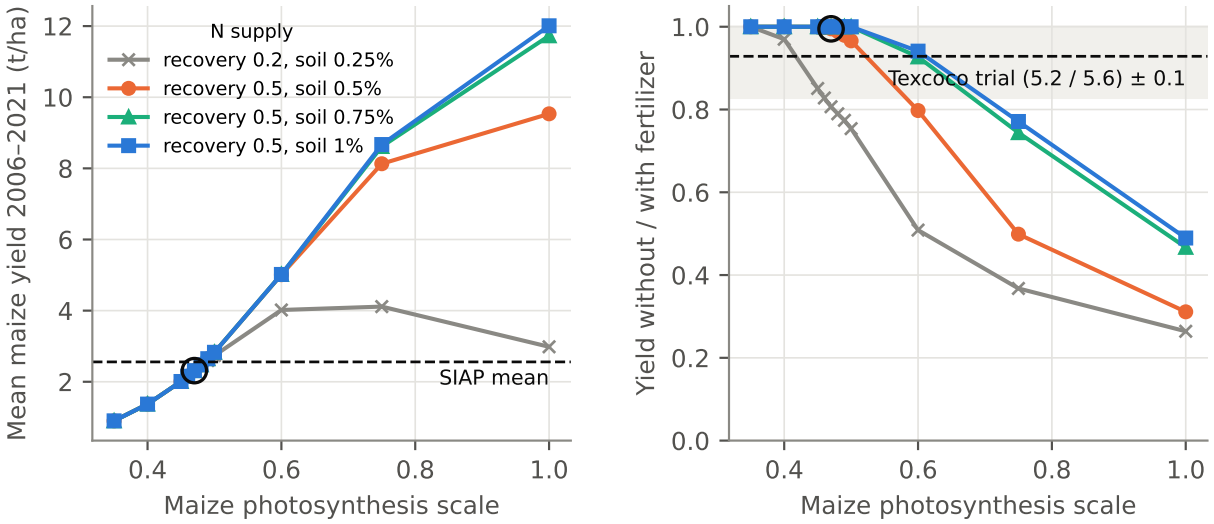


Figure 4: The maize step of the search at the chosen phenology (×0.8). Left: mean maize yield; the dashed line is the SIAP mean. Right: yield without fertilizer divided by yield with fertilizer; the shaded band is the trial's 0.93 ± 0.1. The circle is the chosen candidate. Only low photosynthesis reaches the farm level while keeping the trial's small fertilizer response.

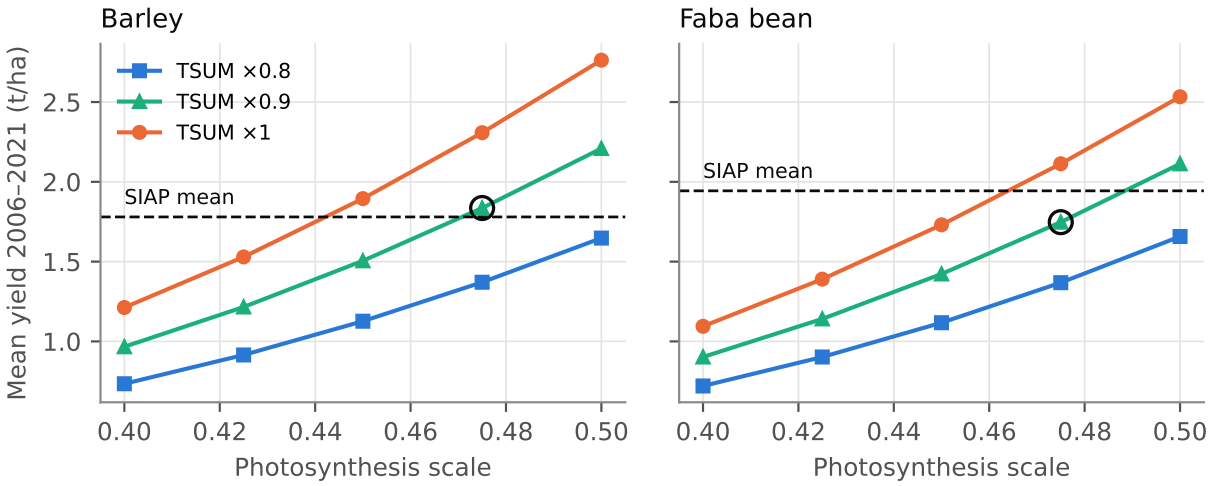


Figure 5: The barley and faba bean steps (fertilizer N recovery 0.5). Each line is a phenology scale; the circled point was chosen; the dashed line is the SIAP mean.

9. What this means for the rotation results

The calibrated model reaches the farm level by lowering photosynthesis, not nitrogen, so nitrogen hardly limits calibrated maize:

Mean yield, all years 2006–2021 (t/ha)	Default model		Calibrated model	
	with fertilizer	without	with fertilizer	without
Maize	6.47	1.92	2.31	2.30
Barley	6.15	3.90	1.81	1.81
Faba bean	9.81	9.77	1.77	1.77

As a result, calibrated maize after barley or after faba bean (2.32 t/ha) yields the same as

maize after maize (2.31 t/ha). The rotation benefits reported with the default model (EXPLORATION.md) need a second look too:

- In the baseline, maize–faba bean gave +0.63 t/ha over maize every year (2006–2025). That advantage comes entirely from the 2023 continuous-maize failure of Section 5 (+12.7 t/ha that year); without 2023 it is −0.003 t/ha.
- With 20% less rain and half fertilizer, the +1.0 t/ha advantage is real in the default model (+0.94 over 2006–2021) and comes through nitrogen left by the faba bean. In the calibrated model nitrogen hardly limits maize, so that benefit would likely shrink (not simulated here).

If the Texcoco trial is right that nitrogen barely limits maize there, the simulated rotation benefits are overstated. If farms are in fact more nitrogen-limited than the station, the first calibration was closer. SIAP cannot tell which, so this needs field data on both fertilizer response and rotations.

10. Limits you should know about

1. One trial decides the nitrogen split. Without the trial constraint, the best maize candidate cut nitrogen instead (RMSE 0.87), with a no-fertilizer ratio of 0.26. The constraint rests on one research-station experiment, whose soil may be more fertile than the average farm field.

2. Photosynthesis absorbs everything WOFOST does not simulate. Weeds, pests, diseases, plant density and local varieties are all lumped into one number per crop. It gets the level right, but it is not a measurement of any of them.

3. Phenology is not pinned by observations. We have no observed flowering or maturity dates, so phenology was bounded by agronomic judgment rather than fitted to dates. Maize chose the shortest allowed cycle.

4. The observations are coarse. SIAP numbers are municipal estimates, often rounded (2.6, 2.4), and recent barley and faba bean figures describe 4–12 hectares (31 and 18 ha of barley in 2017 and 2018). Faba bean's decline over time is not explained by weather.

5. One point represents the whole municipality. One POWER grid cell and one soil point; real fields differ in soil depth (e.g. *tepetate* layers), sowing date and management. The model's weather is also about 2.5 °C colder than the fields (Section 5).

6. Reproducibility. PCSE re-downloads NASA POWER weather when its cache is older than 90 days, so the saved runs (weather cached on 2026-10-12) cannot be reproduced exactly after about 2027-01-10. `calibration.json` records the weather date, git commit and PCSE version, and `requirements.txt` pins PCSE 6.0.13 because `wofost_rotations.py` patches that version.

11. How to use it and reproduce it

Everything is opt-in; nothing changes for existing commands. From the repository root:

```
# One-at-a-time diagnostics, Table 2 (about 2 min, fit years only)
python -m src.crop_models.calibration --sensitivity
# The search (about 13 min on 8 cores); writes fit_history.csv and fitted_knobs.json
python -m src.crop_models.calibration --fit
# Calibrated baseline -> data/rotations_calibrated/ (other scenarios with --scenarios)
python -m src.crop_models.calibration --simulate
python -m src.crop_models.calibration --simulate --scenarios baseline rain80
# The figures of this document
python docs/technical_report/make_calibration_figures.py
# Unit tests, including tests/test_calibration.py
```

```
python -m unittest discover -s tests
```

Files produced in `data/rotations_calibrated/`:

- `baseline.csv` (and any other scenario requested): the same columns as the default runs, with the scenario named `baseline_calibrated`, so default and calibrated runs can be loaded together;
- `calibration.json`: the knobs, fit and holdout years, versions, weather date and the scores of Table 7;
- `fit_history.csv` and `fitted_knobs.json`: every candidate tried in the search and the chosen knobs (copy them into `CALIBRATED` after a refit).

The farmer notebook still loads the default runs and rescales them with `farm_gap` factors. This compares the calibrated run with SIAP instead:

```
calibrated = load_seasons(["data/rotations_calibrated/baseline.csv"])
farm_gap(calibrated, siap, scenario="baseline_calibrated")
```

12. Glossary

Area weighting Combining simulated maize so that each previous crop counts by the share of maize fields that followed it, from SIAP areas.

Calibration Adjusting a few model parameters so the model's output matches local observations.

Candidate One combination of knob values tried by the search.

Climatology A naive prediction: the average observed yield of the fit years, every year.

Coordinate search Fitting one group of knobs (here, one crop) at a time while keeping the others fixed.

Correlation r Whether simulated and observed good and bad years line up; ignores the level.

Grid search Trying every combination of a short list of values for each knob.

Holdout Years kept out of the search and used only to test the result.

Mineralization Release of plant-available nitrogen from soil organic matter by microbes.

N recovery The share of applied fertilizer nitrogen that the crop can actually take up.

Overfitting Matching the fit years' noise so closely that new years are predicted badly.

Persistence A naive prediction: last year's observed yield.

Phase Where in its sequence a rotation starts; every phase is simulated so every crop appears each year.

Phenology The timing of crop development (emergence, flowering, maturity).

RMSE Root mean square error: the typical size of the difference between simulation and observation, in t/ha.

SIAP Mexico's agri-food and fisheries information service, source of the observed yields.

Skill score $1 - \mathrm{RMSE} \,/\, \mathrm{RMSE}_{\mathrm{climatology}}$: above 0 beats the average, below 0 does worse.

Spin-up The first simulated year, discarded because it starts from assumed soil conditions.

TSUM Thermal time (degree-days) a crop needs to reach flowering (TSUM1) or maturity (TSUM2).

Yield per sown hectare Production divided by the area sown, so lost fields count as zero yield.